

SIMATS ENGINEERING
CSA0310-DATA STRUCTURE
LAB
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Searching Algorithm - Singly Linked List Data Structures

1.Linear Search

```
Enter 1 for manual input, 2 for random: Enter 10 integers:
Enter target value: Target 34 found at position 6
Comparisons made: 6
Status: Found
Time taken: 0.000 ms
```

2. Binary Search

```
Enter 15 numbers: Sorted array: 5 10 15 20 25 30 35 40 45 50 55 60 70 80 90
Enter target: low=0, high=14, mid=7, arr[7]=40
low=8, high=14, mid=11, arr[11]=60
low=8, high=10, mid=9, arr[9]=50
low=8, high=8, mid=8, arr[8]=45
Found at index 8 (1-based: 9)
Comparisons: 4
Array remains sorted: Yes
```

3. Binary Search- Iterative

```
Sample Input 6
Sample Output (after sort):
22 24 28 29 31 32
First >=30: index 4 (31)
Last <=25: index 1 (24)
```

4. Linear Search -Sort

```
120 140 200 250 275 310 500
First >=300: index 5 (310)
Last <=150: index 1 (140)
```

5. Linear Search - Anagrams

```
Anagrams found at positions: 1 2
Matches: listen silent
```

6. Linear Search on Array

```
Comparison Trace:
10 != 40
20 != 40
30 != 40
40 = 40
Target Found at Position 4
Comparisons Made = 4
```

7. Binary Search - Array

```
Found at index 2
```

8. Linear Search -Temperature

```
17 18 19 21 24 25 30  
Last <20: index 2 (19)
```

9. Linear Search -Prices

```
90 150 180 200 500 650 780  
First >=500: index 4 (500)  
Last <=200: index 3 (200)
```

10. Bin. Search-Iterative-Temp

```
22 24 28 29 31 32  
First >=30: index 4 (31)  
Last <=25: index 1 (24)
```